

Creating Vehicular Traffic Simulation with OSS

Here is a tutorial on creating real-time vehicular traffic simulation using SUMO and OpenStreetMap.



SUMO stands for Simulation of Urban Mobility. It is designed as an open source package for microscopic road traffic simulation, while OpenStreetMap is an editable, open source map of the world. This article tells you how to simulate real-time traffic using SUMO and OpenStreetMap.

Prerequisites

I have used SUMO 0.22.0 and Ubuntu 12.04 for this simulation.

For installing SUMO in Ubuntu, follow the steps given below:

1. Download SUMO from <http://sourceforge.net/projects/sumo/files/sumo/>
2. Install some packages that are necessary to build SUMO with a GUI:

```
sudo apt-get install libgdal1-dev proj libxerces-
```

```
c2-dev
```

```
sudo apt-get install libfox-1.6-dev libgl1-mesa-dev libglu1-mesa-dev
```

3. Ubuntu 12.04 does not ship with *libgdal.so*; it ships only with *libgdal1.7.0.so*. So, create a symbolic link:

```
sudo ln -s /usr/lib/libgdal1.7.0.so /usr/lib/libgdal.so
```

4. Now decompress the downloaded file by using the following command:

```
tar xzvf filename.tar.gz
```

5. Enter into the source directory and call *configure*, *make* and *make install*: